From documents to reviewable relationships

A Private Wealth tool for solicitors and advisers preparing an estate-planning or trust review: reconstruct family and trust relationships, inspect the documentary Evidence, and decide what to accept.

Alice Chen —— Parent of ——→ **Bob Chen**

Attendance Note - Meeting with Alice Chen

“Alice Chen confirmed that Alice Chen and David Chen are the parents of Bob Chen.”

A relationship is useful when the reader can inspect the words behind it.

University hackathon case study / English pages 1-20

Synthetic prototype. Simplified Chinese follows on page 21.

Case 01: exact source-backed relationship; editorial diagram.

Reconstruct the facts. Keep the words within reach.

Before an estate-planning or trust review, a Private Wealth solicitor or adviser needs to establish the family relationships, recorded Trust roles and documents supporting each connection. PCG makes that factual map inspectable before advice.

The question inside Case 01

> “The purpose of the meeting was to establish an accurate understanding of the current family and beneficiary position before any further review or substantive advice is undertaken.”

Exact passage · Attendance Note – Meeting with Alice Chen

Across documents, a person may appear under another name and a later note may confirm or change a role. A **Matter** is the review workspace; **Sources** are its documents; **Evidence** is the quoted text behind each relationship. The map preserves the route back to those words.

Built for this hackathon: a relationship workspace, an AI-to-graph pipeline with deterministic checks, and a Workbench for controlled comparisons. The proposed benefit is a clearer factual handover: preparer and reviewer can inspect the same claims and supporting words. Practitioner usefulness remains to be tested.

Reading route Product 3-7 / Method 8-12 / Findings 13-17 / Next 18-20

One person.
Two different claims.

Bob Chen's family connection and his beneficiary status answer different questions. Case 01 states both explicitly. The graph must retain each relationship with its own supporting Evidence; one claim does not establish the other.

Family relationship

Alice Chen ⟶ **Bob Chen** Parent of

> “Alice Chen confirmed that Alice Chen and David Chen are the parents of Bob Chen.”

Trust role

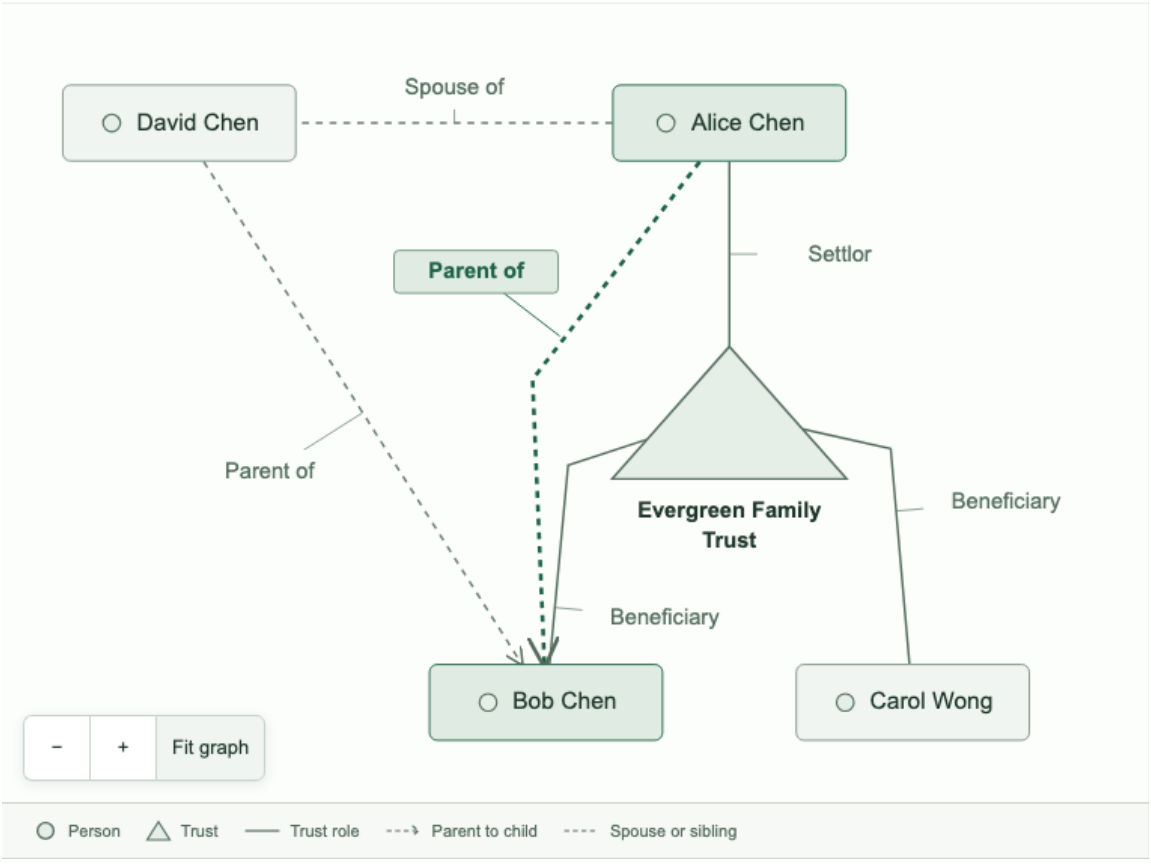
Bob Chen —— **Evergreen Family Trust** Beneficiary of

> “Alice Chen confirmed that Bob Chen is a beneficiary of the Evergreen Family Trust.”

The same note identifies Carol Wong as another beneficiary while recording no family relationship between Carol and the Chen family. A shared Trust connection is therefore not permission to add a family link. Documentary support governs what can be proposed.

A graph you can question

An incoming adviser asks: what supports Bob Chen's connection to this family? Select Alice Chen's parent relationship, inspect the quoted words, then open the Source in context. Bob's beneficiary role needs separate Evidence.



Select the claim

Alice Chen is the parent of Bob Chen. Family relationships and Trust roles remain distinct claims.

The selected connector leads to the quotation below. Compare Bob's separate Trust role on page 3.

Actual practitioner workspace: selected relationship detail

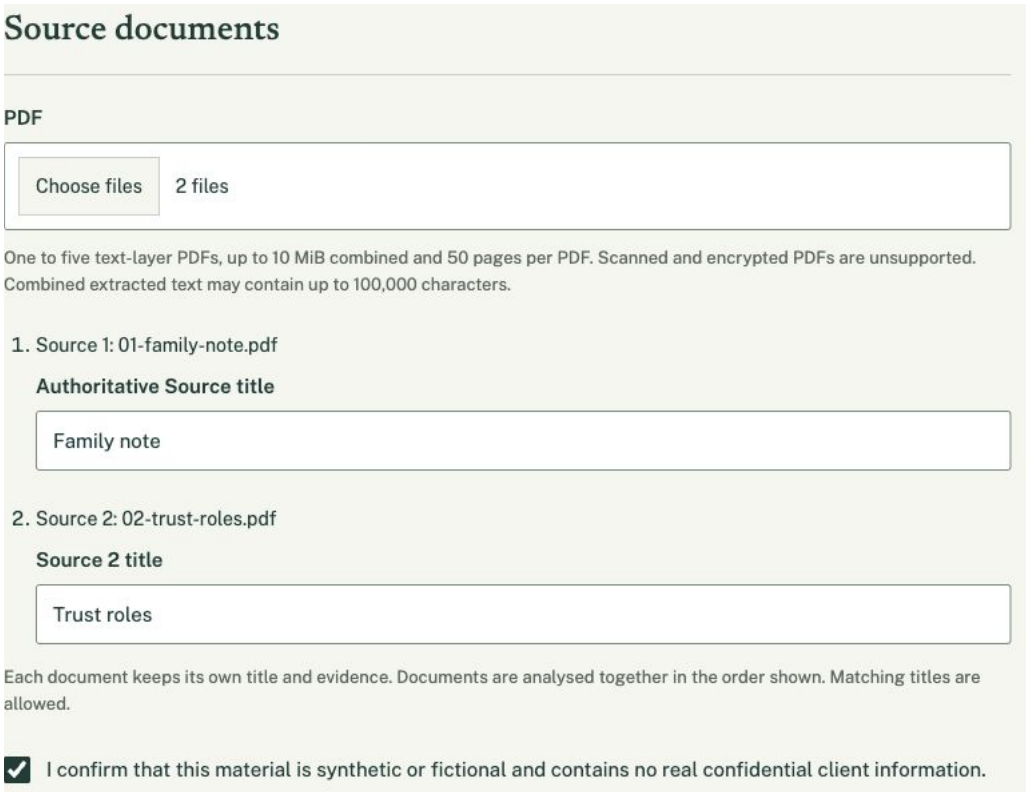
Source detail: Attendance Note - Meeting with Alice Chen

She was then asked about Bob Chen so that the relationship between the individuals being discussed was clear. Alice Chen confirmed that Alice Chen and David Chen are the parents of Bob Chen.

The quotation retains its Source identity. Selecting “View in source” reveals the passage in context; a highlighted sentence supports inspection but does not replace professional judgment.

Several PDFs.
One reviewable proposal.

An allowlisted signed-in user supplies a Matter reference, a title and one to five synthetic text-layer PDFs. Each document keeps its own Source identity and title. The documents are analysed together to produce one combined proposal.



Two distinct Sources

The recorded intake uses “Family note” and “Trust roles”. The Source titles and selected PDFs shown here were entered in the real application.

Historical authenticated UI capture. Synthetic inputs; Google and model boundaries substituted. This records the workflow, not extraction accuracy.

Deterministic PDF acquisition establishes the Authoritative Source text used for extraction, exact Evidence matching and later review. The raw PDFs are discarded after the intake request, including when processing fails; the reviewable record is the extracted text.

A proposal is saved only after successful analysis. It remains proposed state until the professional makes an explicit decision.

PDF acquisition: ADR 0005. Historical capture provenance in product-demo/.

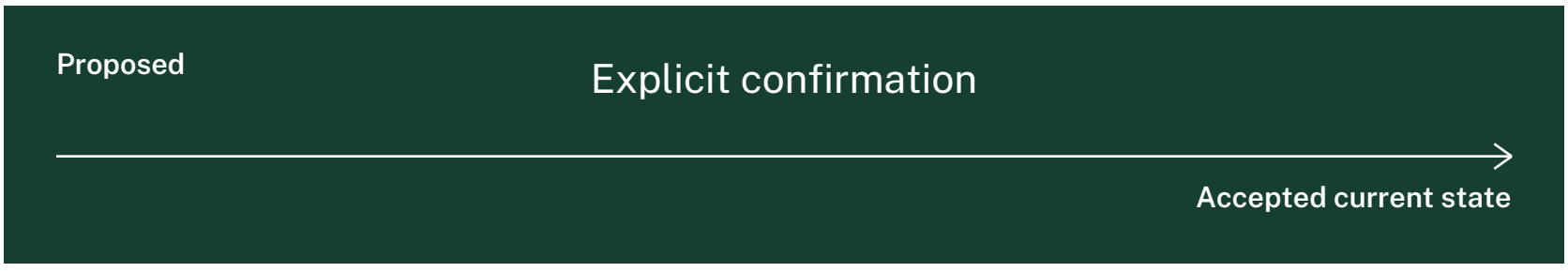
The decision belongs to the professional.

A mechanically valid graph is still a machine proposal. The saved proposal can be reopened for review; it does not silently become a Matter. The professional inspects the complete graph and supplied Sources, then confirms the whole proposal or discards the intake.

Actual whole-proposal decision



Historical authenticated capture, before confirmation. The local model was substituted; proposal confirmation and persistence used the real application paths.



Confirmation records the practitioner's decision to adopt the whole proposed map for the Matter; each claim retains its Sources for further checking. The recorded demonstration saves that state, reopens the Matter and inspects a parent relationship without reanalysis.

This is whole-proposal acceptance. Per-relationship editing and incremental updates to an existing Matter are outside this workflow.

Recorded transaction, confirmation and reopen: product-demo/PROVENANCE.md.

One relationship. More than one Source.

In authored Case 04, Andrew Chan's beneficiary relationship to the Hillcrest Family Trust appears in the first meeting and is confirmed at a follow-up. The relationship can carry Evidence from both Sources, with each quotation attributed to its own text.

Current UI detail: two supporting Sources



Source 2 · sources/note_02.txt · note_02_evidence_001

> The discussion returned first to the beneficiary position. Victoria Chan was asked to confirm the position concerning Andrew Chan. She confirmed that Andrew Chan is a beneficiary of the Hillcrest Family Trust.

Curated authored ideal in the current review workspace. This is synthetic demonstration data, not a model outcome or accepted Matter.

The second note supplements the first rather than revising it. Corroboration here does not demonstrate general conflict resolution, temporal reasoning or automatic updates to an accepted Matter.

Exact follow-up passage and Source identity: authored Case 04.

AI interprets.
Code enforces the rules.

Language requires interpretation. Graph integrity should not depend on another guess. PCG assigns those jobs to different parts of one fixed Extract / Resolve / Construct pipeline.

Extract relationships

Model

Read the ordered Sources together. Propose relationship candidates with named endpoints and Source-attributed quotations.

Resolve names

Configuration choice

Application intake currently uses no resolution. Research can add a constrained model stage to correct supported endpoint names.

Construct the graph

Deterministic code

Validate types, references and exact quote occurrence. Assign stable identities, pool Evidence and produce canonical graph structure.

Mechanical validity is only one boundary

Code can establish that a quotation occurs in a Source. That does not prove it supports the relationship. Professional acceptance remains explicit; deterministic benchmark evaluation runs separately.

Architecture: ADRs 0001 and 0009; current application configuration.

A case-led method, from hypothesis to product

The methodology starts with a documentary challenge. Development examples make the expected behaviour explicit; separate held-out examples test it. The Workbench makes the outcomes inspectable, and the surrounding application supports the workflow.

1

Author a case hypothesis

Define the new difficulty and what to hold stable, so a failure can be diagnosed.

2

Build development test cases

Write synthetic Source samples and explicit ground-truth relationships and Evidence.

3

Build a separate held-out test set

Create new synthetic situations and check their truth separately from authoring.

4

Measure and plot in the Workbench

Compare frozen runs, inspect relationship and Evidence scores, and trace errors to Sources.

5

Build the surrounding application

Support document intake, graph and Evidence inspection, and explicit professional acceptance.

Hackathon scope: baseline extraction plus constrained alias resolution. The roadmap on page 19 sets out the broader programme; it is not a list of completed experiments.

A small benchmark, with an explicit boundary

The first five roadmap scenarios supplied development cases. A separate held-out set tested whether those choices carried into new documentary situations. All cases were authored for this project and contain synthetic information.

10
held-out cases

14
Sources

39
expected relationships

What the held-out cases ask

Held-out 02 — **Dense wording**
Separate sibling status from beneficiary roles.

Held-out 05 — **A completed replacement**
Retain earlier facts; remove the superseded trustee.

Held-out 08 — **Matching document titles**
Preserve distinct Source identities.

Ground truth was checked separately from authoring by AI validators. These were internal checks, not external human or institutional validation. The retained comparison freezes its inputs and records its outcomes; it does not turn this small synthetic set into evidence of field performance.

Benchmark counts: retained publication; independent AI checks: validation records.

Count the relationships. Then check the Evidence.

The primary score compares exact relationship edges: the named starting entity, relationship type and named target entity. Direction matters for directed links; symmetric links are canonicalised. A plausible sentence alone is not a match.

$$F1 = 2 \times TP / (2 \times TP + FP + FN)$$

TP: matched edges · FP: extra edges · FN: missing edges

Two valid averages answer different questions

94.87%

Micro-F1, baseline A. Pool edge counts across all cases. Cases with more relationships contribute more edges.

93.33%

Macro-F1, baseline A. Average the ten case F1 scores. Each case contributes equally, whatever its size.

Provenance is assessed separately. Approved-quote checks concern matched relationships; Source/quote-pair checks concern evidence pairs. Their populations differ, and neither score should be read as another relationship-accuracy percentage.

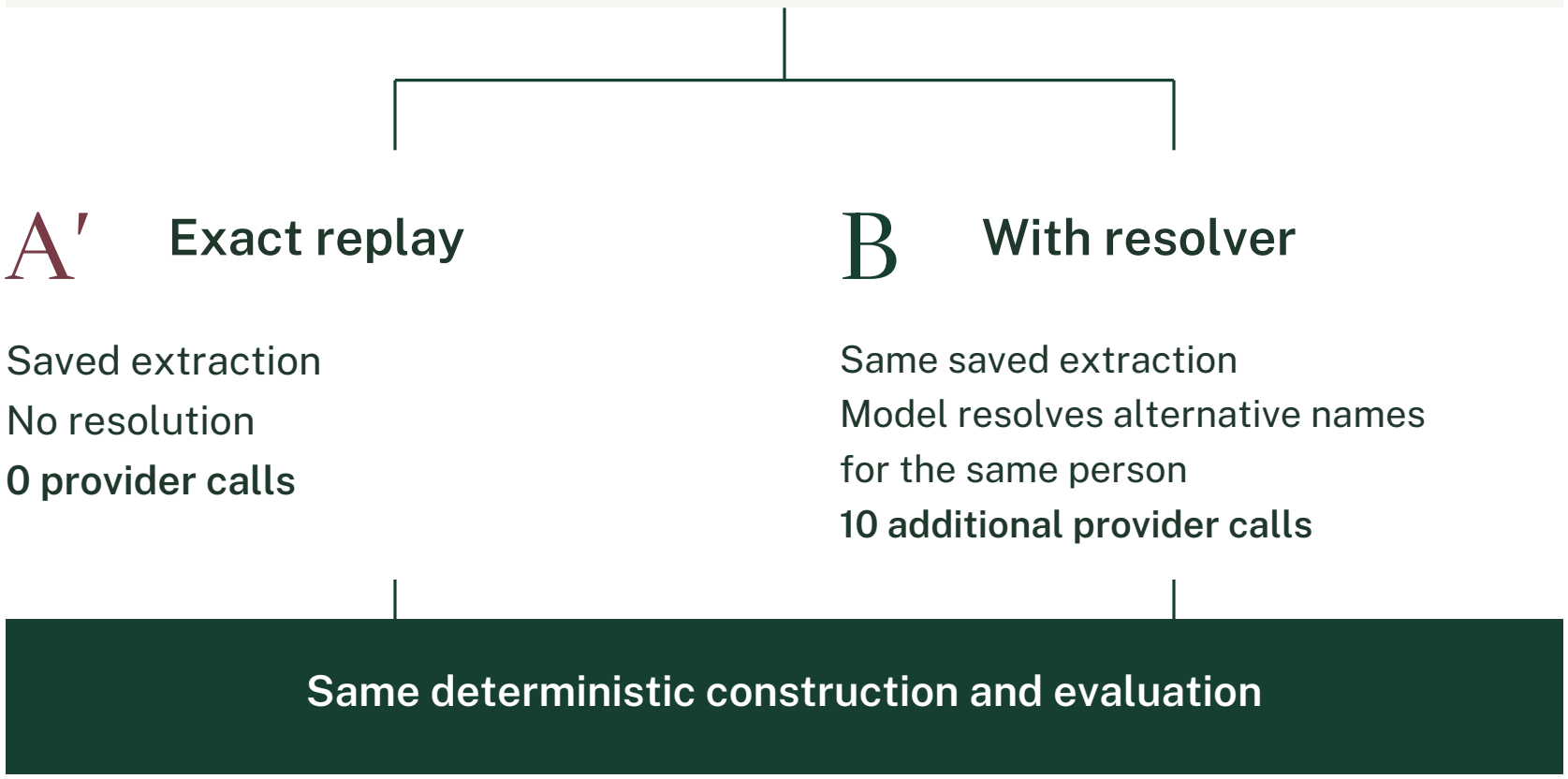
Definitions: semantic_edges_v1; separate approved-provenance diagnostics.

Change one stage. Keep the extraction fixed.

Can a name-only correction recover missed relationships without rewriting the claims? The comparison reuses saved extraction outputs and adds a zero-call replay control, isolating resolution from fresh extraction variability.

A Fresh baseline extraction

10 cases. One fresh model call per case.
Retain all extraction outputs for reuse.



A′ reproduces A's outcomes. This helps isolate the resolver's effect from new extraction variability; it is **not an independent replication**. A′ and B use the same implementation and frozen inputs.

Retained A / A′ / B records; post-hoc exploratory comparison.

One person.
Two corrected endpoints.

Heldout_03: factual interview note

> “Beatrice Ng attended the factual interview and asked to be called Bea. In this note, Bea means Beatrice Ng; no second person is being introduced.”

Exact Source excerpt from the synthetic held-out case. Editorial figure.

Endpoint correction			Unchanged target
Settlor of			
Bea	⟶	Beatrice Ng	Larch Family Trust

“Bea is the settlor of the Larch Family Trust.”

Parent of			
Bea	⟶	Beatrice Ng	Isaac Ng

“Bea is the parent of Isaac Ng.”

The already-correct Isaac Ng beneficiary relationship remains unchanged.

The resolver returns endpoint-name edits only. Code preserves candidate count, relationship types, directions and quotations. Replacement names must occur in the Sources or extracted endpoints. Whether two names identify the same person remains a model judgment.

Two corrections explain the entire gain

On 10 project-authored synthetic held-out cases, relationship micro-F1 rose from **94.87% to 100%**. The change came from two endpoint-name corrections for one person in one case.



Actual published Workbench chart. Run 1 = A; Run 2 = A′; Run 3 = B. The full 0-1 F1 scale is preserved; points are not a time trend.

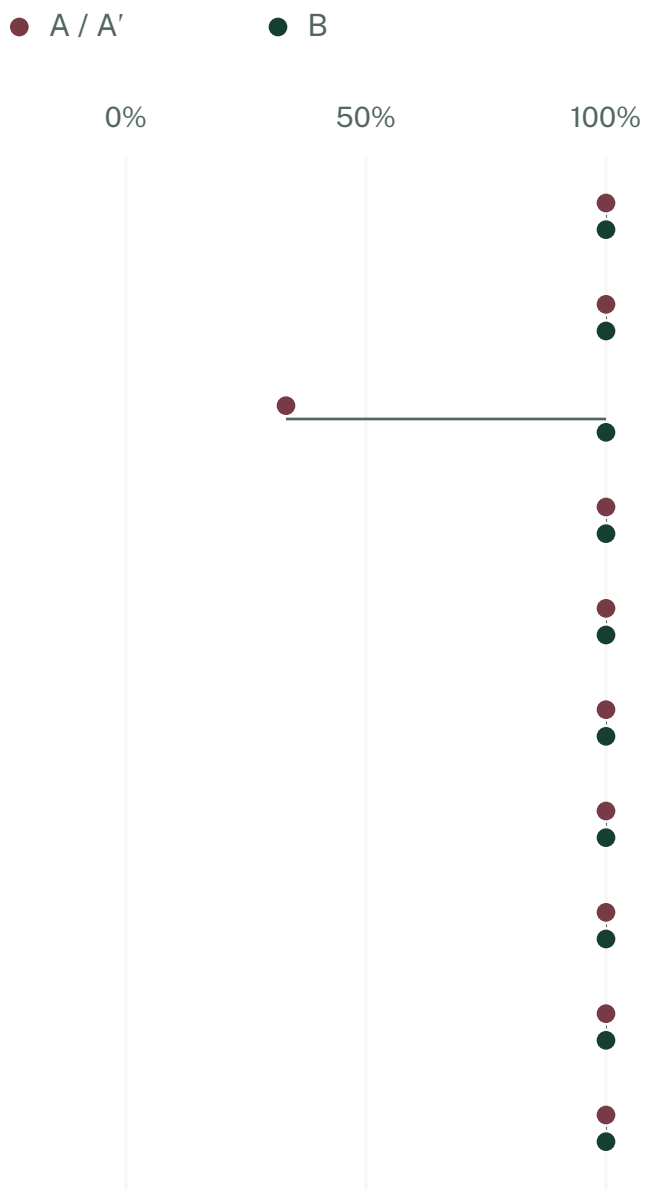
Pooled across 39 expected relationships	A / A′	B
Matched relationships	37	39
Extra / missing relationships	2 / 2	0 / 0

Evidence checks remained imperfect: B passed 36/39 approved-quote checks and 37/41 Source/quote-pair checks, using separate denominators. The resolver is implemented in the research pipeline; application intake currently uses baseline extraction. Small, synthetic, post-hoc comparison.

Nine cases already had perfect relationship F1

Every case except the explicit Bea short form scored 100% relationship F1 before resolution and after it. This locates the gain precisely; it does not demonstrate a broad improvement across ten distinct failures.

Relationship F1 per case



Case 03: 33.33% to 100%. Points use the full 0–100% scale. The paired markers are separated vertically so equal scores remain visible. A′ exactly reproduces A. One execution per case; no repeated-trial uncertainty estimate.

Case names and plotted F1 values: retained publication. Editorial chart.

Perfect edges do not mean perfect Evidence

B passed **36/39 approved-quote checks** and **37/41 Source/quote-pair checks**. These are separate populations. Exact approved boundaries can fail even when a shorter quotation supports the relationship.

Case 02: Grace Wong is a sibling of Hugo Wong

Approved span

“Felix Wong is the settlor of the Amber Family Trust, and Grace Wong and Hugo Wong are siblings and are both beneficiaries of the Amber Family Trust.”

Returned span · same Source, later start boundary

“Grace Wong and Hugo Wong are siblings and are both beneficiaries of the Amber Family Trust.”

Exact retained excerpts. This mismatch does not imply an unsupported sibling claim.

The incremental resolver usage

10 calls · 9,821 tokens

9,432 input + 389 output. The 3,456 cached input tokens are included, not additional. USD cost is unpriced in the retained record.

All calls use deepseek-flash. This post-hoc comparison uses one provider/model and one execution per case; it supplies no repeated-trial estimate of stability or end-to-end latency claim.

A useful finding.
A defined boundary.

The experiment demonstrates a traceable correction under controlled inputs. It also shows why an attractive graph and a high relationship score must remain open to inspection.

What the record supports

The resolver repaired the two observed endpoint names without changing the other nine cases' relationship outcomes. The retained examples also include multi-source corroboration, identical Source titles and explicit trustee replacement. These are observations within this dataset.

What it cannot establish

Ten short, project-authored synthetic cases are not a representative client-document collection. The study used one model, deepseek-flash, with one extraction and one resolver execution per case. There were no repeated trials or uncertainty estimates. The comparison was post-hoc; general identity resolution and real-document reliability remain untested.

An inspectable record, not a fresh model promise

Frozen Sources, settings, code identities, outputs, checksums and usage preserve the comparison. Replaying saved outputs reproduces the record, not a fresh model response. Workbench approval binds a frozen plan to a shared call budget; uncertain attempts remain counted and are not automatically retried.

The next version needs new evidence

The hackathon established baseline extraction and a constrained resolver. Continue the twenty-scenario programme by isolating new failure modes and testing improvements, alongside the product work below.

Validate the review workflow

Observe practitioners finding missing claims, checking Evidence and deciding whether to accept a proposal. Test the Trust layout and measure task outcomes before claiming time savings or usability.

Support change over time

Design incremental Matter updates, reviewed changes, corrections and historical state. A narrow trustee-replacement example does not yet provide a general temporal or contradiction-handling system.

Design the real-data boundary

Develop Matter authorization, tenancy and ownership, and explicit security, privacy, retention and data-handling requirements. Current sign-in and allowlisted actions do not authorize real client material.

Research should use newly unseen cases, richer documents and repeated runs. The published held-out set is now visible and must not be treated as unseen evidence in subsequent work informed by it.

Twenty scenarios. One staged research programme.

We planned a progression of hypotheses about extraction: introduce a specific difficulty, diagnose the failure, then test an improvement. Within the hackathon window, we established the **baseline and a constrained alias resolver**. The wider programme remains ahead.

Planned challenges, in increasing complexity

01–05 have authored development cases. 06–20 are planned scenarios, not validated capabilities.

01 Explicit family / Trust facts

02 Dense facts and overlapping roles

03 Aliases for the same person

04 Cross-document reconciliation

05 Changed relationships

06 Contradictory Sources

07 Ambiguous role attribution

08 Indirect relationship descriptions

09 Negation and hypothetical claims

10 Larger, denser family graphs

11 Multiple Trusts

12 Corporate / non-person trustees

13 Different document genres

14 Locally resolvable pronouns

15 Sibling / cousin distinction

16 Overlapping Evidence spans

17 Longer Sources and distractors

18 Unnamed participants / omission

19 Historical versus current state

20 Combined stress test

---

20 roadmap scenarios · 5 authored development cases · 10 separate held-out cases. These are different scopes. The held-out study evaluates the baseline-plus-resolver slice; it does not complete the twenty-scenario roadmap.

Case progression: cases/README.md. Scenario labels condensed for this document.

Make every relationship reviewable

For Private Wealth advice, begin with a family-and-trust map whose connections can be checked against the documents. This university prototype connects synthetic PDF intake, graph review and explicit acceptance, backed by an inspectable baseline-plus-resolver comparison.

Try the public review workflow

private-client-graph-production.up.railway.app

Open a synthetic Matter, select a relationship, inspect its Evidence, then view the passage in its Source. Creating and confirming synthetic intake requires an allowlisted account.

Inspect the retained research

Research results and downloadable original publication

Read the implementation

github.com/alistairkung/private-client-graph

Sources and visual provenance

Product and architecture

README, glossary and ADRs 0001, 0005, 0006, 0009-0011.

Benchmark and retained results

Held-out cases, validator reports and the frozen A / A′ / B publication.

Product images

PR #160 current UI; PR #155 historical workflow captures. Synthetic fixtures; model substituted.

Research chart and editorial figures

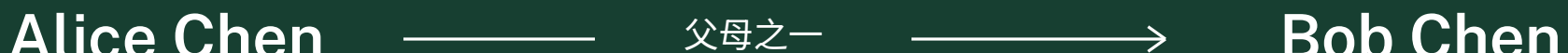
Actual production chart captured 18 Oct 2026. Other diagrams explain recorded facts.

Synthetic-data prototype. Not validated for real client use.

Links and current product inspected on 18 October 2026.

从文档出发
让关系可供审阅

为准备遗产规划或信托审查的律师及顾问打造的私人财富工具：重建家庭与信托关系，核查文档中的证据引文，并决定确认哪些内容。



会面记录 — 与 Alice Chen 会面（英文原文）

“Alice Chen confirmed that Alice Chen and David Chen are the parents of Bob Chen.”

读者能查阅关系背后的原文，关系图才有实用价值。

大学黑客松案例研究 / 中文版第21–40页

合成数据原型。英文版见第1–20页。

案例01：关系由准确原文支持；此图为说明性示意图。

重建事实。
随时查阅原文。

开展遗产规划或信托审查前，私人财富律师或顾问需要厘清家庭关系、文档记载的信托角色，以及支持每项联系的文档。Private Client Graph 让这张事实关系图在提供建议前即可接受核查。

案例01中的问题

> “The purpose of the meeting was to establish an accurate understanding of the current family and beneficiary position before any further review or substantive advice is undertaken.”

英文原文 · 会面记录 — 与 Alice Chen 会面

同一个人在不同文档中可能使用不同姓名，后续记录也可能确认或变更其角色。**事项（Matter）**是审阅工作区；**来源文档**是其中的文档；**证据引文**是支持每项关系的原文摘录。关系图保留通往这些原文的查证路径。

本次黑客松的成果：关系审阅工作区、带确定性校验的 AI 图谱生成流水线，以及用于受控比较的研究工作台。预期价值是让事实交接更清晰：准备人与审阅人能够核查相同的关系主张及支持原文。其对专业人员的实用性仍待验证。

阅读路线 产品23–27页 / 方法28–32页 / 发现33–37页 / 下一步38–40页

同一个人。
两项不同的主张。

Bob Chen 的家庭关系与受益人身份回答的是不同问题。案例01明确记载了两者。图谱须分别保留每项关系及其证据引文；一项主张不能证明另一项。

家庭关系

Alice Chen ⟶ **Bob Chen** 父母之一

> “Alice Chen confirmed that Alice Chen and David Chen are the parents of Bob Chen.”

信托角色

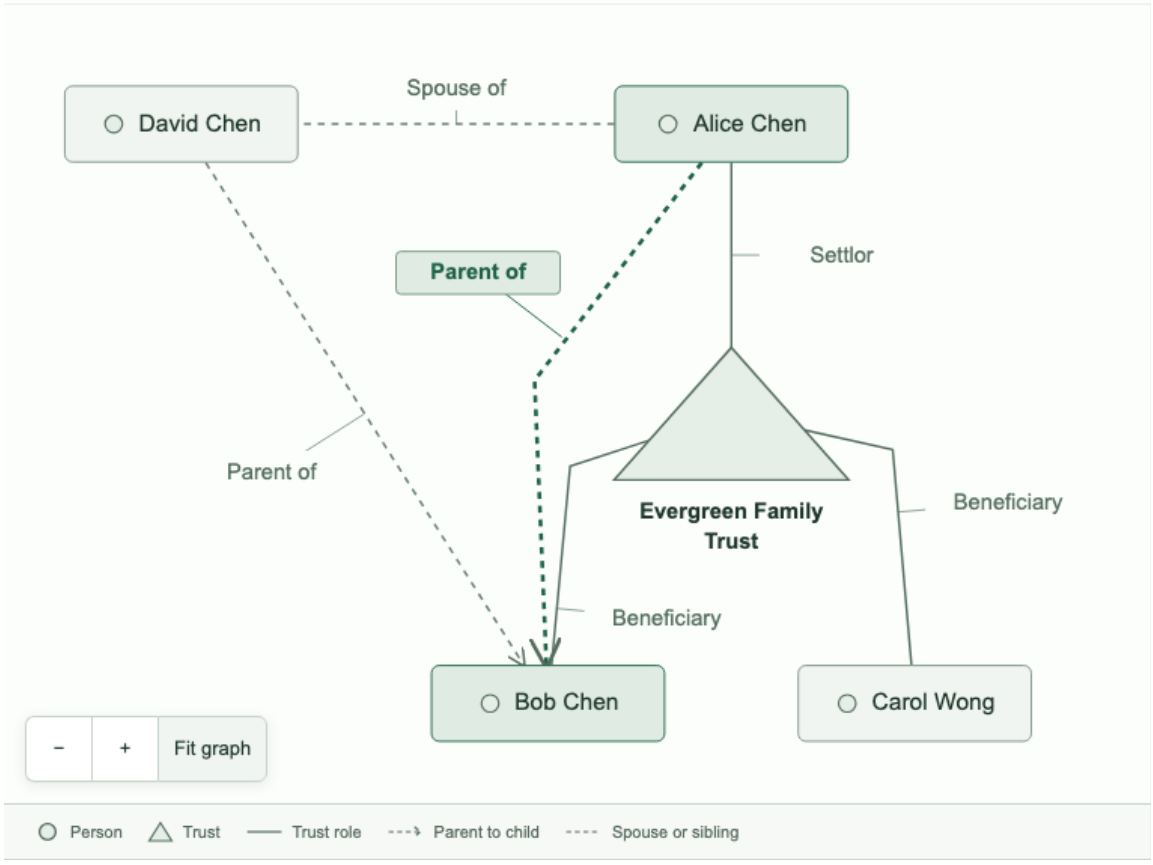
Bob Chen —— **Evergreen Family Trust** 是其受益人

> “Alice Chen confirmed that Bob Chen is a beneficiary of the Evergreen Family Trust.”

同一份记录还将 Carol Wong 列为另一位受益人，但未记载 Carol 与 Chen 家族存在家庭关系。因此，共同关联某项信托，并不足以添加家庭关系。能够提出哪些关系，取决于文档支持。

一张经得起追问的图

接手的顾问会问：什么证据支持 Bob Chen 与这个家庭的联系？选中 Alice Chen 的父母关系，核查引文，再打开来源文档查看上下文。Bob 的受益人角色需要独立的证据引文。



选择关系主张

Alice Chen 是 Bob Chen 的父母之一。家庭关系与信托角色仍是各自独立的主张。

选中的连线指向下方引文。Bob 的另一项信托角色见第23页。

实际专业人员工作区：所选关系详情

来源文档详情：会面记录 — 与 Alice Chen 会面（英文原文）

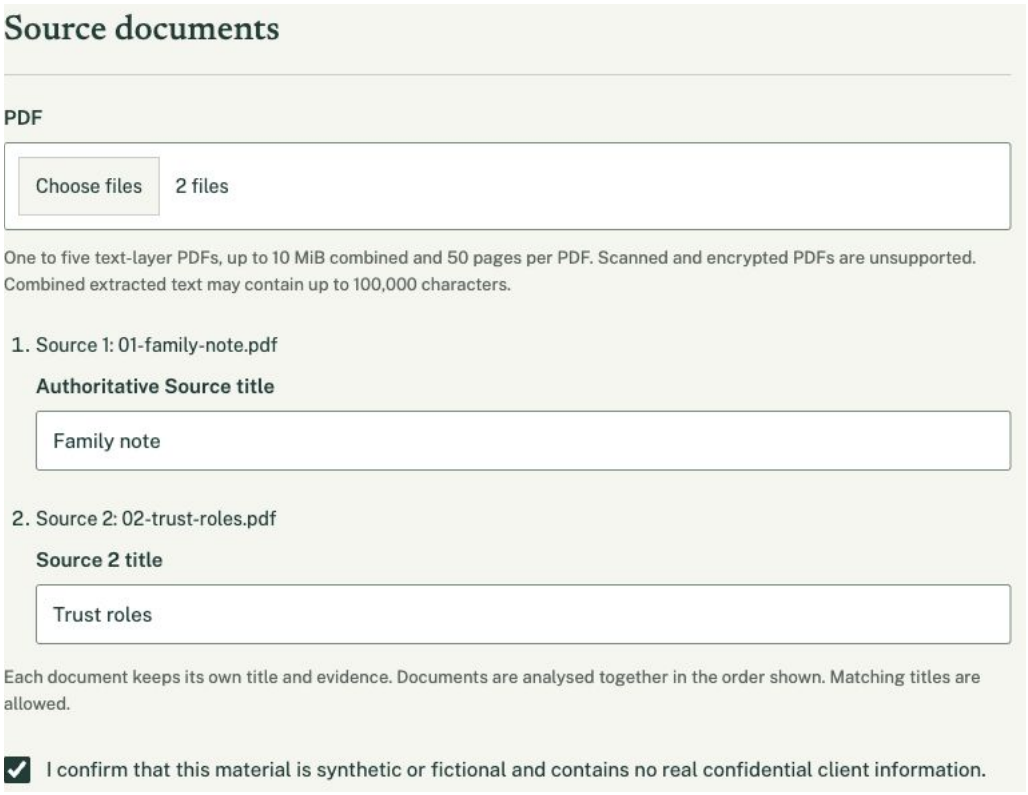
She was then asked about Bob Chen so that the relationship between the individuals being discussed was clear. Alice Chen confirmed that Alice Chen and David Chen are the parents of Bob Chen.

引文保留其来源文档标识。选择“查看来源文档”可显示该段落的上下文；高亮句子有助于核查，但不能替代专业判断。

当前界面截图；合成数据，模型调用由测试实现替代。PR #160。

多份 PDF。
一份可审阅的方案。

白名单内已登录的用户提供事项编号、标题及一至五份含文本层的合成 PDF。每份文档保留独立的来源标识与标题。系统共同分析这些文档，生成一份合并的待审方案。



两份独立的来源文档

录制中使用的文档名为“Family note”和“Trust roles”。这些标题与所选 PDF 均在实际应用中录入。

此前录制的已认证界面。输入为合成数据；Google 认证与模型调用使用替代实现。该记录展示操作流程，不代表提取准确率。

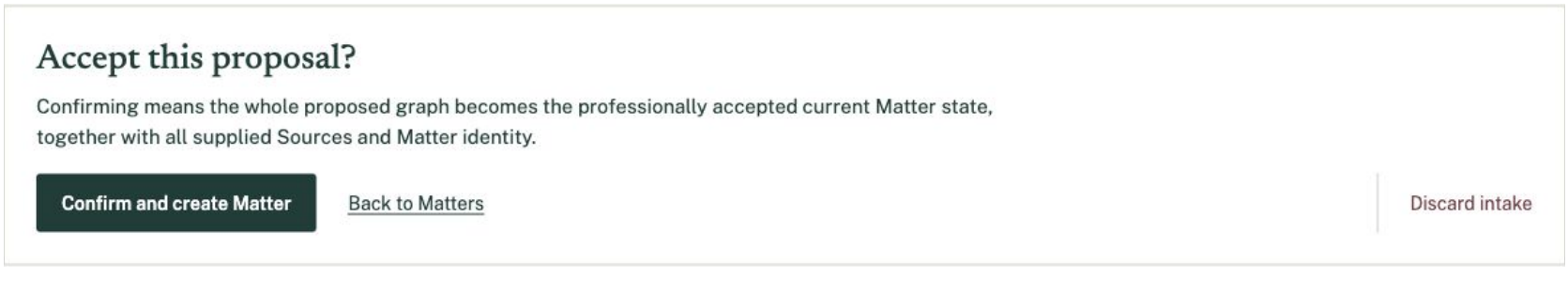
确定性的 PDF 获取流程生成权威来源文本，供信息提取、证据引文精确匹配和后续审阅使用。导入请求结束后即丢弃原始 PDF，处理失败时亦然；供审阅的留存记录是提取出的文本。

仅在分析成功后保存待审方案。专业人员明确作出决定前，方案始终处于待确认状态。

决定权
属于专业人员。

通过机械校验的图谱仍只是机器提出的方案。已保存的方案可重新打开审阅，不会悄然成为事项。专业人员核查完整图谱及提交的来源文档后，确认整个方案，或放弃本次导入。

实际的整份方案确认操作



此前录制的已认证界面，处于确认之前。本地模型使用替代实现；方案确认与持久化则通过真实应用流程完成。



确认操作记录专业人员将整份关系图采纳为该事项当前状态的决定；每项主张仍保留其来源文档，供进一步核查。录制演示保存了该状态，重新打开事项，并在无需再次分析的情况下查看父母关系。

这里确认的是整份方案。逐项编辑关系及对现有事项作增量更新，均不在此流程范围内。

已录制的事务、确认与重新打开流程：product-demo/PROVENANCE.md。

同一项关系。
多份来源文档。

在项目编写的案例04中，Andrew Chan 作为 Hillcrest Family Trust 受益人的关系，首次见于初次会面记录，并在后续会面中得到确认。该关系可同时携带两份来源文档的证据引文，且每条引文均标明对应文本。

当前界面详情：两份支持文档



来源文档2 · sources/note_02.txt · note_02_evidence_001

> The discussion returned first to the beneficiary position. Victoria Chan was asked to confirm the position concerning Andrew Chan. She confirmed that Andrew Chan is a beneficiary of the Hillcrest Family Trust.

当前审阅工作区展示项目编写并整理的理想结果。这是合成演示数据，并非模型输出或已确认的事项。

第二份记录补充了第一份，并未修订它。此处的相互佐证不代表系统具备通用冲突消解、时间推理或自动更新已确认事项的能力。

后续会面的英文原文及来源文档标识：项目编写的案例04。

AI 理解语义。代码执行规则。

语言需要理解，图谱完整性则不应依赖另一次猜测。Private Client Graph 在固定的“提取 / 归一 / 构建”流水线中，将两项职责分配给不同环节。

提取关系

模型

将按序排列的来源文档共同阅读，提出候选关系，包含两端实体名称及注明来源文档的引文。

姓名归一

可选配置

应用导入目前不执行姓名归一。研究可加入受约束的模型环节，校正有证据支持的关系端点名称。

构建图谱

确定性代码

校验类型、引用及引文是否逐字出现在原文中。分配稳定标识、汇集证据引文，并生成规范化图谱结构。

机械有效性只是其中一道边界

代码能够确认引文存在于来源文档中，却不能据此证明它支持该关系。仍须由专业人员明确确认；确定性的基准评估则独立运行。

架构：ADR 0001、0009；当前应用配置。

以案例为导向，从假设走向产品

方法从文档解读难题出发。开发案例明确预期行为，独立的留出测试案例检验这些行为。研究工作台让结果可供检查，配套应用则支持整个工作流程。

1

提出案例假设

明确新增难点及保持不变的条件，以便诊断失败原因。

2

构建开发测试案例

编写合成来源文档样本，明确关系与证据引文的标准答案。

3

构建独立留出测试集

创设新的合成情境，并将标准答案核查与案例编写分开进行。

4

在研究工作台中测量与绘图

比较冻结的运行记录，检查关系与证据引文得分，并回溯来源文档定位错误。

5

构建配套应用

支持文档接收、图谱与证据引文检查，以及专业人员的明确确认。

黑客松范围：基线提取加受约束的别名解析。第39页的路线图列出更广泛的研究计划，并非已完成的实验清单。

作者的方法；案例假设、留出测试记录与研究工作台实现。

小规模基准，明确适用边界

路线图前五个场景提供了开发案例。独立留出测试集检验这些设计选择能否适用于新的文档情境。所有案例均为本项目编写，信息全部为合成内容。

10

留出测试案例

14

来源文档

39

预期关系

留出测试案例检验什么

留出测试 02

密集表述

区分兄弟姐妹关系与受益人角色。

留出测试 05

已完成的更替

保留先前事实；移除已被替换的受托人。

留出测试 08

相同的文档标题

保留各来源文档的独立身份。

AI 核查者独立于案例编写过程核查标准答案。这些属于内部检查，并非外部人员或机构的验证。留存的比较冻结输入并记录结果，但这不能使小规模合成测试集成为实际场景表现的证据。

基准数量：留存的发布记录；独立 AI 检查：验证记录。

先核对关系，再检查证据引文

主要得分精确比较关系边：具名起始实体、关系类型和具名目标实体。有向关系须匹配方向；对称关系则作规范化处理。仅有看似合理的句子并不算匹配。

$$F1 = 2 \times TP / (2 \times TP + FP + FN)$$

TP：匹配的边 · FP：多余的边 · FN：遗漏的边

两种有效的平均方式回答不同问题

94.87%

微平均 F1，基线 A。汇总所有案例的关系边计数。关系较多的案例贡献更多关系边。

93.33%

宏平均 F1，基线 A。取十个案例 F1 得分的平均值。无论规模大小，每个案例的权重相同。

溯源质量单独评估。核准引文检查针对已匹配的关系；来源文档／引文配对检查针对证据配对。两者的统计对象不同，均不应解读为另一种关系准确率。

定义：semantic_edges_v1；另行进行核准溯源诊断。

只改变一个阶段，固定提取结果

仅修正名称，能否在不改写陈述的情况下找回遗漏关系？比较复用已保存的提取输出，并加入零调用的重放对照，将解析效果与重新提取的波动分开。

A

重新执行基线提取

10个案例，每例重新调用模型一次。
保留全部提取输出以供复用。

A′

精确重放

已保存的提取结果
不进行解析
0次服务商调用

B

加入解析器

复用同一提取结果
模型解析同一人的
不同名称
额外10次服务商调用

相同的确定性构建与评估

A′ 复现了 A 的结果。这有助于区分解析器的效果与重新提取的波动，但**并非独立复现实验**。A′ 与 B 使用同一实现和冻结的输入。

留存的 A / A′ / B 记录；事后探索性比较。

同一个人，修正两个端点

Heldout_03：事实访谈记录

> “Beatrice Ng attended the factual interview and asked to be called Bea. In this note, Bea means Beatrice Ng; no second person is being introduced.”

合成留出测试案例的来源文档英文原文节选。说明示意图。

端点修正

目标实体不变

委托人关系

Bea ⟶ Beatrice Ng — Larch Family Trust

“Bea is the settlor of the Larch Family Trust.”

父母之一

Bea ⟶ Beatrice Ng — Isaac Ng

“Bea is the parent of Isaac Ng.”

已正确识别的 Isaac Ng 受益人关系保持不变。

解析器仅返回端点名称修改。代码保留候选关系数量、关系类型、方向和引文。替换名称必须出现在来源文档或已提取的端点中。两个名称是否指向同一人，仍由模型判断。

全部提升，来自两处修正

在本项目编写的10个合成留出测试案例上，关系微平均 F1 从**94.87%升至100%**。变化来自一个案例中同一人的两处端点名称修正。



实际发布的研究工作台图表。运行1 = A；运行2 = A′；运行3 = B。保留完整的0-1 F1 刻度；数据点不代表时间趋势。

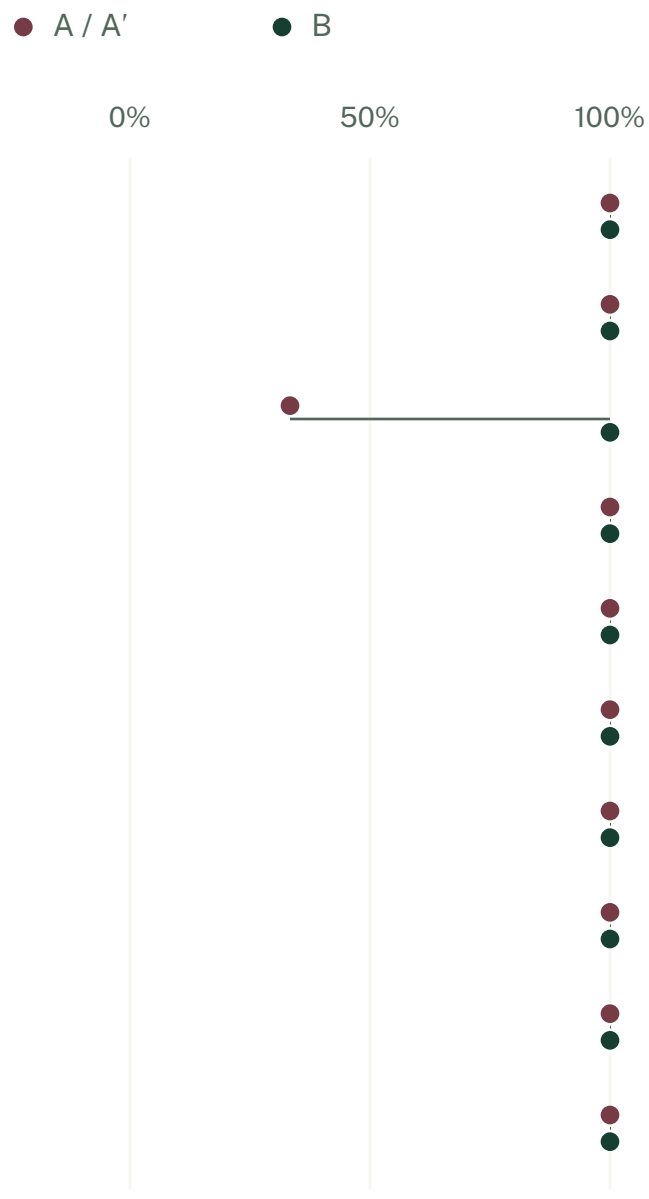
汇总39条预期关系	A / A′	B
匹配的关系	37	39
多余／遗漏的关系	2 / 2	0 / 0

证据引文检查仍未全数通过：B 通过36/39项核准引文检查和37/41项来源文档／引文配对检查，两者分母不同。解析器已在研究流水线中实现；应用的文档接收目前使用基线提取。这是一项小规模、合成数据、事后比较。

九个案例的关系 F1 原已达到满分

除明确使用 Bea 简称的案例外，其余案例在解析前后的关系 F1 均为100%。这精确定位了提升来源，但未证明对十种不同失败均有普遍改善。

各案例的关系 F1



案例03：33.33%升至100%。数据点采用完整的0–100%刻度。配对标记在垂直方向错开，确保相同得分仍然可见。A′ 精确复现 A。每例仅执行一次，未以重复试验估计不确定性。

案例名称与图示 F1 值：留存的发布记录。说明图表。

关系边满分，证据引文仍有缺口

B 通过**36/39项核准引文检查**和**37/41项来源文档／引文配对检查**。两者统计对象不同。即使较短引文支持该关系，也可能因不符合核准片段的精确边界而未通过检查。

案例02：Grace Wong 与 Hugo Wong 为兄弟姐妹

核准片段

“Felix Wong is the settlor of the Amber Family Trust, and Grace Wong and Hugo Wong are siblings and are both beneficiaries of the Amber Family Trust.”

返回片段 · 同一来源文档，起点较后

“Grace Wong and Hugo Wong are siblings and are both beneficiaries of the Amber Family Trust.”

留存的英文原文节选。此项不匹配并不意味着兄弟姐妹关系缺乏支持。

解析器的增量用量

10次调用 · 9,821 tokens

9,432输入 + 389输出。其中已含3,456个缓存输入 tokens，并非额外用量。留存记录未计算美元费用。

全部调用均使用 deepseek-flash。这项事后比较只使用一个服务商／模型，每例执行一次；未通过重复试验估计稳定性，也未声称端到端延迟表现。

精确的核准／预测片段及 B 的增量用量：留存的发布记录。

有价值的发现，有明确的边界

实验展示了在受控输入下可追溯的修正，也说明为何即使图谱清晰、关系得分很高，仍须允许进一步检查。

记录支持的结论

解析器修复了观察到的两个端点名称，其他九个案例的关系结果未变。留存案例还涵盖多文档佐证、相同的来源文档标题及明确的受托人更替。这些均是本数据集内的观察。

尚不能确立的结论

本项目编写的十个简短合成案例，并不能代表客户文档集合。研究只使用 deepseek-flash 一个模型，每例执行一次提取和一次解析。没有重复试验或不确定性估计。比较为事后进行；通用身份解析与真实文档可靠性仍未测试。

可核查的留存记录

冻结的来源文档、设置、代码标识、输出、校验和与用量保存了这次比较。重放已保存输出可复现记录，并不意味着新的模型调用会产生相同响应。研究工作台的审批将冻结计划与共享调用预算绑定；结果不确定的尝试仍计入用量，且不会自动重试。

下一版本，需要新的证据

黑客松完成了基线提取与受约束的解析器。后续应继续二十场景研究计划，隔离新的失败模式并测试改进，同时推进以下产品工作。

验证审阅流程

观察专业人士如何发现遗漏陈述、检查证据引文及决定是否确认待审方案。测试信托布局并衡量任务结果，再提出节时或易用性方面的主张。

支持随时间变化

设计事项的增量更新、经审阅的变更、更正及历史状态。一个范围有限的受托人更替案例，尚不足以构成通用的时序或矛盾处理系统。

设计真实数据边界

制定事项授权、租户隔离与归属机制，并明确安全、隐私、留存及数据处理要求。现有登录与白名单操作并不授权使用真实客户材料。

研究应采用从未接触过的新案例、更丰富的文档与重复运行。已发布的留出测试集现已公开，后续研究若参考了该集合，就不能再将其视为未见过的证据。

二十个场景，分阶段推进研究

我们规划了一系列逐步推进的提取假设：引入特定难点、诊断失败，再检验改进。在黑客松期间，我们建立了**基线与受约束的别名解析器**。更广泛的计划仍有待开展。

计划中的难题，复杂度逐步递增

01–05已编写开发案例。06–20为规划场景，尚非经验证的能力。

01 明确的家庭／信托事实

02 密集事实与角色重叠

03 同一人的别名

04 跨文档核对整合

05 关系变更

06 来源文档互相矛盾

07 角色归属不明确

08 间接描述的关系

09 否定与假设性陈述

10 更大、更密集的家庭图谱

11 多个信托

12 公司／非自然人受托人

13 不同文档体裁

14 可据局部语境解析的代词

15 区分兄弟姐妹与堂表亲

16 证据引文片段重叠

17 更长的来源文档与干扰信息

18 未具名参与者／遗漏

19 历史与当前状态

20 综合压力测试

20个路线图场景 · 5个已编写的开发案例 · 10个独立留出测试案例。三者范围不同。留出测试研究评估的是基线加解析器这一部分，尚未完成二十场景路线图。

案例进阶：cases/README.md。本文对场景名称作了精简。

让每一条关系
都经得起审阅

私人财富咨询可从一张家庭与信托图谱开始，其中的关系均可对照文档核查。这个大学项目原型贯通合成 PDF 接收、图谱审阅与明确确认，并以可检查的基线加解析器比较提供支持。

体验公开审阅流程

private-client-graph-production.up.railway.app

打开一个合成事项，选择关系，检查证据引文，再查看来源文档中的对应段落。创建与确认合成文档接收需要白名单账户。

检查留存研究

研究结果与可下载的原始发布文件

阅读实现代码

github.com/alistairkung/private-client-graph

资料来源与视觉内容溯源

产品与架构

README、术语表及 ADR 0001、0005、0006、0009-0011。

基准与留存结果

留出测试案例、验证器报告及冻结的 A / A′ / B 发布记录。

产品图像

PR #160：当前界面；PR #155：历史流程截图。使用合成测试数据，并替换了模型。

研究图表与说明示意图

实际生产环境图表截于2026年10月18日。其他示意图用于说明已记录的事实。

合成数据原型。未经真实客户使用验证。